

Scaling for Cultural Equity: Investigating the Impact of Training Data Size on Cultural Bias in Vision-Language Models

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Abstract

Vision language models (VLMs) are expanding rapidly, with real-world applications ranging from multimodal search to decision-making systems. However, these models often inherit and amplify cultural biases. While larger datasets generally enhance model performance, they also reinforce dominant cultural biases. Understanding how the volume of training data influences these biases is critical to developing equitable AI systems. Although existing research on the impact of scaling datasets focuses predominantly on performance metrics and the prevalence of harmful content, the impact of scaling datasets on cultural bias in VLMs remains unexplored. This paper investigates whether increasing the volume of fine-tuning data will negatively impact model performance in interpreting cultural concepts, leading to increased cultural bias. Our findings will reveal the trade-offs between scaled datasets, cultural bias mitigation, and model generalization, and inform strategies for training VLMs that are culturally inclusive.

1 Introduction

AI development has rapidly become a global phenomenon with advancements outside the United States, such as DeepSeek (China) and DeepMind (UK). With the rapid integration of NLP and language model systems in the global audience, it is imperative that the systems respond in culturally sensitive manners, and generalize well to their vast audience, especially if the application is user-facing (Pawar et al., 2024).

Recent advancement in VLM capabilities ranges from image captioning, text-to-image generation, to multimodal classification. The emergence of models such as DALL-E, StableDiffusion, and others, as documented in recent studies, marks a significant evolution in the field, although many of these models are developed under restricted conditions, limiting broader public scrutiny and independent

evaluation on data quality (Birhane et al., 2023). As the performance of VLMs has consistently improved over time, it is increasingly crucial to evaluate these systems against more rigorous standards (Figure 4). VLMs should be able to understand and represent diverse cultural values—including beliefs, rituals, and traditions (Liu et al., 2021).

When we refer to cultural bias, it is more often than not referring to the predominant Western-centric data and the alienation of Non-Western cultures. This definition and perspective of cultural bias in the context of language models has been adopted in various papers (Ye et al., 2024; Ananthram et al., 2025; Liu et al., 2025; Pouget et al., 2024; Yadav et al., 2025; Liu, 2024). Although research on cultural bias in large language models (LLMs) has been extensively covered, we will focus on bias and cultural bias for VLMs specifically.

Existing work on VLMs and bias predominantly address societal biases such as gender and race, (Chen and Dou, 2024); whereas, others have examined the effects of pre-training language distribution to reveal VLMs’ disproportional Western bias (Ananthram et al., 2025). Even noticeable datasets for training VLMs, such as LAION Schuhmann et al. (2022), are sourced from Western languages and cultures (Pawar et al., 2024). Evaluating existing VLMs using a custom multimodal benchmarking tool supports the notion of lower performance in VLMs in recognizing Non-Western cultural cues and context (Liu et al., 2025).

The “scaling laws” of training data in machine learning is often justified by the prevailing narrative that “scale beats noise,” where larger datasets are assumed to compensate for imperfect annotations (in the case of LLMs) or noisy samples (Kaplan et al., 2020; Schuhmann et al., 2021). This practice is believed to enhance generalization and improve coverage of rare concepts. Supporting this view, Jia et al. (2021) argue that web-scale datasets, despite their inherent noise, can bypass heavy data

curation while achieving better performance. However, emerging evidence complicates this narrative. While scaling may boost average performance on VLM downstream tasks, it amplifies the societal biases encoded in the data. [Birhane et al. \(2023\)](#) noted that increasing the size of the LAION dataset from 400M to 2B examples not only exacerbated racial bias—such as the association of Black individuals with criminality—but also led to a 12% rise in hate speech prevalence.

The unexplored dimension in this discourse is the nuanced interplay between data scaling and cultural bias — particularly how models’ understanding of culturally-specific concepts degrades or evolves as training data expands. Existing evaluations prioritize performance metrics over cultural sensitivity, failing to assess the negative consequences of perpetuating harmful stereotypes ([Naous et al., 2024](#); [Yang et al., 2024](#)).

2 Related Work

Recent studies reveal that the integration of multiple modalities in VLMs holds significant promise for enhancing value understanding through visual cues, yet it also introduces new risks ([Ye et al., 2024](#); [Sahili et al., 2025](#)). In particular, multi-modal systems can exhibit substantial variability in aligning with diverse cultural contexts, sometimes amplifying societal and historical stereotypes far beyond the levels observed in unimodal models ([Yadav et al., 2025](#)).

Many recent benchmarks have laid a foundational framework for evaluating cultural understanding in VLMs, yet each exhibits its own limitations. [Liu et al. \(2025\)](#) introduced CultureVLM and CultureVerse, a comprehensive benchmark that spans nearly 20,000 cultural concepts and dimensions, but exclusively focus on the recognition of static concepts (Figure 5). Similarly, CulturalVQA employed image-question pairs to probe cultural facets across 11 countries, revealing significant regional performance disparities ([Nayak et al., 2024](#)). [Li et al., 2024](#) leveraged an LLM-powered framework to simulate cross-cultural dialogues, generating extensive synthetic data that improved alignment with Hofstede’s cultural dimensions with CulturePark. Lastly, GlobalRG, by [Bhatia et al., 2024](#), extended evaluations to both universal and culture-specific concepts across 50 countries on image retrieval tasks.

Our work builds upon these existing cultural

benchmarks by extending the evaluation of VLMs to investigate the impact of model scale on cultural understanding. While prior studies have established geo-diverse benchmarks, they haven’t systematically explored how scaling laws—typically studied in model performance across downstream tasks—interact with cultural nuance comprehension. Our research bridges this gap by assessing VLM performance across varying model sizes on a self-curated image-caption dataset derived from Wikipedia. This approach allows us to analyze whether larger models exhibit improved cultural understanding or if biases persist regardless of scale.

3 Methodology

Our methodology consists of four phases: 1) Fine-tune open-source VLMs on training datasets of varying sizes; 2) Curate benchmark of cultural images and corresponding captions by web scraping Wikipedia; 3) Evaluate model performance with cultural questions and answers; 4) Analyze and compare results across models to assess their performance on culturally nuanced concepts.

3.1 Models

For the experiment, open-source VLMs including LLaVA (7B), OpenCLIP (ViT-B-32), and SmolVLM (2B) were used. These models, each with fewer than 8B parameters, are more manageable within computational constraints. LLaVA and OpenCLIP are well-known for their strong multi-modal capabilities, making them effective for tasks that involve image-text relationships. Initially, MobileVLM v2 (1.7B) was considered, but it was replaced by SmolVLM due to its better compatibility with the evaluation process. SmolVLM is a small and an efficient model that also offers optimal memory usage, making it an efficient choice for resource-constrained environments. Fine-tuning smaller pre-trained models ensures clearer insights, as larger models may overshadow the effects of fine-tuning. Consequently, this ensures that any observed performance differences are likely due to the volume of fine-tuning data rather than the overwhelming influence of large-scale pre-training corpus.

Since our project focuses on the cultural understanding of models with varying training data sizes, we will fine-tune each model using subsets of the LAION-400M dataset. We will benchmark the base models and fine-tuned models using our

Wikipedia evaluation dataset with culturally specific questions and images to evaluate their ability to interpret diverse cultural concepts correctly.

3.2 Datasets

For fine-tuning, we will be using LAION-400M, which contains 400M large-scale, multimodal data, including multilingual image-text pairs, that will provide the models with cultural context. We chose to use LAION-400M because LAION is open-source and curated for machine learning research purposes, thereby reducing preprocessing workload. Using random sampling, we will subsample from the LAION-400M dataset to ensure that samples are feasible for fine-tuning, given the computational and resource constraints of our project. We will randomly sample 0.05%, 0.1%, 0.5%, and 0.1% of the data from the whole dataset (Figure 6).

To evaluate the performance of our base models and fine-tuned models, we will curate a Wikipedia image-based benchmark of culturally-nuanced questions and answers through web scraping. This benchmark will help us assess the models’ understanding and representation of cultural concepts amongst the different training dataset scales.

To construct our evaluation dataset, we define a curated list of Wikipedia category pages to scrape for image-caption pairs (e.g., Culture of India, Japanese Holidays, French Weddings, etc.). First, a robust HTTP session will be established using Requests and Urllib3 with built-in retry logic for reliable access. Next, we retrieve the designated Wikipedia category page and use BeautifulSoup to parse its HTML content, extracting individual page links from specific group elements. For each article link, its corresponding content are fetched and employed with BeautifulSoup to locate relevant images, identified by a particular tag containing both the image URL and its caption. The captured elements are then compiled into a structured JSON format, and the aggregated data is saved to an output file.

4 Experimental Setup

We investigated the following research question:

Do varying training dataset sizes impact vision-language models’ performance when processing culturally nuanced concepts?

Specifically, we examine whether increasing the volume of training data leads to a reduction, an

increase, or a more complex effect of cultural bias within these models.

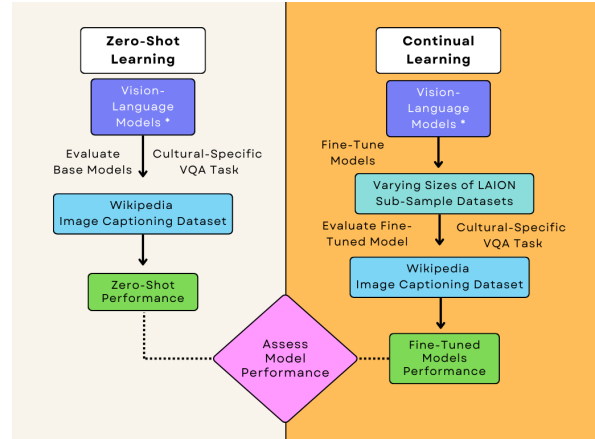


Figure 1: Experimental Design: Zero-Shot Learning vs. Continual Learning for Cultural Equity in Vision-Language Models

To investigate the impact of training data size on cultural bias within vision-language models, we employ a two-pronged experimental design (Figure 1). First, we establish a zero-shot learning baseline by evaluating pre-trained VLMs directly on a culturally-specific Visual Question Answering (VQA) task using our Wikipedia Image Captioning Dataset. Second, we fine-tune these pre-trained models on our subsampled data from LAION-400M. Following fine-tuning, these models are evaluated on the same cultural-specific VQA tasks. Finally, we compare the performance of the zero-shot and fine-tuned models across different volumes of fine-tuning data, employing statistical tests to determine the significance of observed differences in performance and bias.

4.1 Model Evaluation

To assess potential cultural bias in our models, we will employ VQA to test and compare the performance. A randomly selected subset of images from our Wikipedia dataset will be fed into our models, along with a question about the image’s contents and multiple choice answers. These answers will include the correct “ground truth” caption and distractor captions derived from other images. To directly compare bias across cultures, we will analyze performance across various groupings, such as Western vs Non-Western cultures or specific continent groupings. We will then calculate the group-specific accuracy of the model’s ability to select the correct answer.

5 Experimental Process

5.1 Preparing Fine-Tuning Data

Downloading the LAION-400M dataset involved several steps. First, the dataset was downloaded onto HPC as parquet files, utilizing the 50TB of storage available on the scratch directory. A connection to Hugging Face was established and the data, though compressed in hundreds of parquet files, were downloaded in four batches. A Python script was used to share URLs and captions to extract images from the parquet files, enabling parallel processing and downloading. `img2Dataset` was used to convert the URLs into images and save the captions. Each download shard contained thousands of .tar files with 10,000 image-text pairs, totaling 65M out of the 400M image text pairs, as downloading the full dataset was computationally expensive.

5.2 Downloading Models

The process for downloading the models to the HPC proceeded smoothly. After installing the appropriate packages for each of the three models, we utilized the `.from_pretrained` function to download them. The downloaded models (from HuggingFace) are as follows: CLIP132M, LLaVA7B, and SMOL2.3B. A model (containing weights from `model.safetensors` files) and processor (along with a tokenizer to process image and text inputs) were required for each of the three. Following the successful download, these models were fine-tuned and evaluated using a Wikipedia-based cultural evaluation dataset, which was assembled through web-scraping.

5.3 Fine-Tuning

Throughout our fine-tuning process, several challenges were encountered due to limited resources. The unavailability of GPUs and insufficient memory allocation restricted the experiment. With the resources allocated to our project, we performed fine-tuning on OpenCLIP on 0.05% and 0.1% subsets of the LAION dataset. In the OpenCLIP fine-tuning process, `CLIPProcessor` and `CLIPModel` were used to handle model tokenization and image normalization. Then, parameter-efficient fine-tuning (PEFT) was performed using LoRA, freezing unnecessary weights, which resulted in training only 1% of the parameters. The dataset was shuffled and images were mapped to its corresponding caption and preprocessed before being loaded into

the model. The model was trained to minimize the loss function. The pairwise similarity between images and captions was computed during the forward pass for each batch and the LoRA adapter weights were updated (with other parameters frozen) during back propagation. The adapter weights were then saved as the fine-tuned model and evaluated with the Wikipedia benchmark. This process was intended for all other subsets of the LAION dataset. Many SLURM jobs were submitted to run the fine-tuning process. Although fine-tuning scripts for LLaVA and SmolVLM were developed, resource constraints on HPC prevented their completion.

5.4 Wikipedia Benchmark Creation

To systematically evaluate our VLMs on culturally diverse concepts, we constructed a Wikipedia-based image-caption benchmark through targeted web scraping. Our objective was to curate a lightweight, structured evaluation dataset composed of a broad range of cultural artifacts and contexts image-caption pairs (Figure 7).

We targeted both Western and Non-Western categories that represent a range of architectural traditions and cultural phenomena, as per our hypothesis. According to our literature review, we also suspect that models perform worse answering cultural concepts for Africa and Asia, therefore, we define a set of continent categories. These categories were chosen to ensure the benchmark covers a diverse span of cultural, historical, and stylistic traditions.

To extract data, we implemented a web scraping pipeline using the `Requests` and `BeautifulSoup` libraries. For each selected category page, we retrieved all linked Wikipedia articles listed under that category. We extracted the image source URL and the associated caption text for each figure. All extracted image-caption pairs were stored in a structured JSON format, where each entry contains fields for the image’s URL and caption. This format ensures compatibility with downstream zero-shot evaluation pipelines.

5.5 Model Evaluation using Wikipedia Benchmark

The models are tasked with selecting the correct answer based on an image and 4 multiple choice options describing the object in the image. The evaluation process included pre-processing the images by fetching them from URLs and applying a vision language pipeline for model inference. Different mod-

els required distinct approaches due to their architecture and prep-processing requirements. OpenCLIP utilized a simple CLIPProcessor and CLIP-Model for handling both the text and image inputs, while LLaVA involved more complex configurations, including HuggingFace Transformer’s “image-text-to-text” pipeline that integrates vision and language tasks, and quantization through the “BitsandBytesConfig” module. SmolVLM, a more lightweight model, also used a pipeline designed for vision-language tasks. All models were tested on questions designed to classify images into cultural categories including Western, Non-Western, and continental groupings. Group-specific accuracies were computed (dividing the number of correct predictions by the total predictions for each group) to assess model performance across these diverse cultural contexts. The results, including overall accuracy and group-specific performance were recorded and saved for hypothesis testing and further analysis.

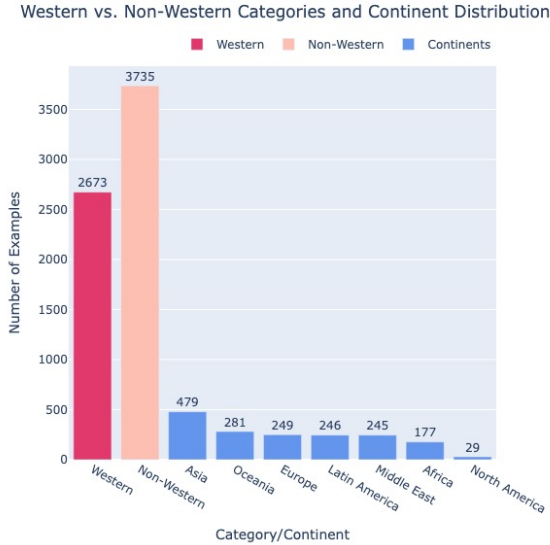


Figure 2: Distribution of Western, Non-Western, and Continents Categories in Wikipedia Benchmark

5.6 Limitations

Resource limitations of HPC impacted the experiment, preventing the completion of fine-tuning for LLaVA and SmolVLM. As a result, the fine-tuning process for OpenCLIP was prioritized (0.05% and 0.1% of the LAION dataset), focusing on comparing the zero-shot and fine-tuned evaluation with the Wikipedia dataset. Several issues arose during the process, including limited availability of powerful GPUs, leading to extensive training times. Debug-

ging was challenging due to frequent “killed” messages, indicating out-of-memory errors. CUDA memory issues also occurred when using higher batch sizes, but reducing the batch size meant that models trained slower as a tradeoff. Consequently, most of our hypothesis testing and results are based on the zero-shot evaluation of the three models.

6 Results

Model Accuracies by Western vs Non-Western

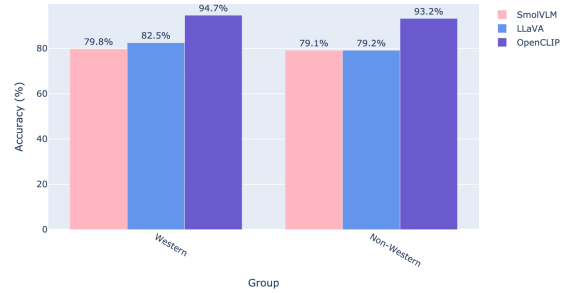


Figure 3: Model Accuracies for Western vs Non-Western for SmolVLM, LLaVA, and OpenCLIP

In figures 3 and 8, the accuracies for SmolVLM, LLaVA, and OpenCLIP are displayed for Western and Non-Western regions, as well as the continents (Table 3). The overall average accuracy is displayed in Table 1. Out of the three models, OpenCLIP outperforms the others in all groups. To quantify bias, we calculated accuracy disparity (the difference in accuracy between the best-performing and worst-performing cultural groups). The accuracy disparity for the models in Western vs Non-Western and continents in figure 9 shows that in the first group, LLaVA has the highest disparity at 3.35, and in the second group, SmolVLM has the highest disparity at 15.5.

6.1 Fine-Tuned OpenCLIP VLM on 0.05% and 0.1%

After fine-tuning OpenCLIP on 0.05% and 0.1% of the LAION dataset, the new model was evaluated with the Wikipedia benchmark, revealing that fine-tuning on these subsets of the dataset provides no difference in cultural bias of the model. The accuracies of the fine-tuned models were the same for each group and the total accuracy. As a result, hypothesis testing was not performed between the zero-shot evaluation and the fine-tuned 0.05% and 0.1% evaluation of OpenCLIP.

Model	Zero-Shot Accuracy	0.1% Fine-Tuned Accuracy	0.05% Fine-Tuned Accuracy
SmolVLM (2B)	79.41%		
LLaVA v1.5 (7B)	80.57%		
OpenCLIP (ViT-B-32)	93.84%	93.84%	93.84%

Table 1: Zero-Shot and Fine-Tuned Evaluation Accuracies for Each Model

Comparison	Test Statistic	P-value	Significant?
SmolVLM: Western vs Non-Western	0.6560	0.5118	No
SmolVLM: Continents	11.9749	0.0625	No
LLaVA: Western vs Non-Western	3.3366	0.0008	Yes
LLaVA: Continents	15.0409	0.0199	Yes
OpenCLIP: Western vs Non-Western	2.3341	0.0196	Yes
OpenCLIP: Continents	7.3994	0.2855	No

Table 2: Z-Test and Chi Squared Test Results On Zero-Shot Evaluation

6.2 Hypothesis Testing

After running our baseline models through the Wikipedia benchmark, statistical tests were conducted to determine whether the performance differences across models are significant between groupings, indicating the presence of cultural bias.

We used a two-proportion z-test for pairwise comparisons between Western and Non-Western groups, as well as between continents. Despite North America having fewer than 30 samples, we consistently applied the z-test for all continent pairs to maintain methodological consistency. For comparisons across all continents, we used a chi-squared test. In all cases, the null hypothesis assumed no difference in accuracy between groups.

The statistical test results largely align with our initial hypotheses and prior literature. LLaVA showed a significant accuracy gap both between Western and Non-Western regions and across continents, consistently favoring Western regions and achieving higher accuracy for Oceania, Europe, and the Middle East. OpenCLIP displayed a significant difference between Western and Non-Western regions, but not consistently across continents, suggesting a more limited but still present Western bias. For SmolVLM, there is not a significant difference in accuracy for both groups being compared, revealing that there is not significant cultural bias in the zero-shot evaluation of SmolVLM.

With the z-test and chi squared test results for all three models (Table 2), there is only a significant difference in accuracies of both groups in LLaVA and in Western vs Non-Western for Open-

CLIP. These experiments revealed that the baseline model of LLaVA and OpenCLIP are culturally biased towards Western culture and SmolVLM is balanced in all groups.

The results are plotted for the continent comparison z-test for SmolVLM (Figure 10), LLaVA (Figure 11), and OpenCLIP (Figure 12) with a significant cutoff at 0.05.

- SmolVLM showed five significant differences, with three involving North America. In these cases (North America vs. Europe, Oceania, and Africa), SmolVLM’s accuracy for North America was the lowest, suggesting that SmolVLM may be least culturally biased toward North American data.
- LLaVA had eight significant pairwise comparisons, but no single continent consistently stood out across comparisons.
- OpenCLIP exhibited a significant difference only between Europe and Asia, indicating a slight bias toward European examples. This finding is consistent with the Western vs. Non-Western bias observed for OpenCLIP.

In addition, expected calibration error (ECE) was performed for OpenCLIP per group (Figure 13). Probability data for each answer in the multiple-choice questions (MCQs) was collected from this model. ECE measures how well the model’s confidence matches the actual accuracy by bucketing predictions into confidence intervals. For each bucket, the ECE is a weighted average of the difference between confidence and accuracy for each

group. Lower ECEs indicate better calibration in models with a good model typically having an ECE below 0.05. For OpenCLIP, the highest ECE was in North America at 0.072 and the lowest was in Europe at 0.0139. In addition, the Non-Western ECE is higher than the Western ECE, providing evidence that OpenCLIP has cultural bias towards Western culture meaning that the training data may be tailored more towards Western regions.

6.3 Qualitative Analysis of Zero-Shot Evaluation

In Table 1, the average total accuracy of each model is displayed. The accuracies of both LLaVA and SmolVLM averaged around 80% while OpenCLIP averaged around 94%. The accuracy disparity between Western and Non-Western groups are 0.68%, 3.35%, and 1.43% for SmolVLM, LLaVA and OpenCLIP respectively. From the results, it is visible that OpenCLIP has the highest overall performance and is the most balanced between continents. LLaVA and SmolVLM perform similarly in terms of accuracy but LLaVA has the largest imbalance between Western and Non-Western, while SmolVLM has the largest imbalance between continents.

OpenCLIP has extremely high and consistent accuracy across all groups. While the z-test revealed that there is a significant difference between Western and Non-Western, OpenCLIP is still robust in typically less-represented continents like Africa. This could be because OpenCLIP is trained on massive, highly diverse image-text datasets and excels at matching images to semantically closest text, regardless of cultural origin. Also, the multiple-choice format in this dataset likely rewards simple semantic matching which is OpenCLIP’s strength.

LLaVA has relatively high accuracy but is less consistent in its performance, performing stronger in Western regions. The model also drops in accuracy for North America and Latin America, compared to the previously mentioned top three continents. This could be due to LLaVA being trained for visual instruction-following, not just caption matching. It particularly excels when reasoning over complex visual scenes or nuanced prompts is needed, but the slight performance drops suggest that the model overthinks simpler image-text matching tasks or misinterprets visual cultural nuances.

SmolVLM also has relatively high accuracy, but

generally performs below the other two models. It exhibits a small gap between Western and Non-Western regions, 0.68, showing that SmolVLM has good generalization across cultures. In addition, the model is particularly strong with data regarding Oceania, Africa, and Asia providing evidence that there is less cultural bias in SmolVLM. This is probably a result of the model being lighter with only 2B parameters, meaning that a smaller capacity could limit its fine-grained reasoning and semantic discrimination. SmolVLM does not “overthink” simple tasks the way LLaVA does, but it also does not match the massive training diversity of OpenCLIP.

6.4 Analysis of Fine-Tuning Outcomes for OpenCLIP

After fine-tuning OpenCLIP on 0.05% and 0.1% of our LAION-400M dataset, the Wikipedia evaluation accuracies stayed the same. A potential explanation for this phenomenon is because of the mismatch between our fine-tuning data and our evaluation data. Fine-tuning the model with the captions reinforces alignment and not discriminative reasoning. MCQs require the model to parse and understand the question, evaluate multiple answers in the context of the image, and select the best one. In contrast, image-caption pairs may not be the best way to fine-tune for this method of evaluation as the model did not learn to rank multiple choice options or deal with structured questions during fine-tuning.

Another possible explanation for these results is that OpenCLIP’s core capability may not be for question-answering tasks. OpenCLIP is a contrastive model, meaning that it learns which text matches to which image rather than which answer choice is correct in MCQs. This fundamental limitation makes OpenCLIP a poor fit for zero-shot QA without specifically adapting to with training or fine-tuning QA data. Our results may be due to our fine-tuning dataset size. While we were constrained by the HPC resources, we were only able to implement our two smallest dataset sizes for fine-tuning OpenCLIP so even though the entire LAION dataset is large, our 0.05% and 0.1% are relatively small and may not have impacted the model enough, especially since the data is not in the question-answer format.

Lastly, this could be a result of the format of our fine-tuning data. Our fine-tuning process did

not incorporate a VQA-style dataset, and without task-format alignment, the model was not able to interpret MCQs and the structure of choosing an option based on a question and image presented. As a result, OpenCLIP is still working from a caption-matching framework, even after fine-tuning. In conclusion, the fine-tuning did not have an impact on the cultural bias present in the OpenCLIP VLM.

7 Conclusion

Our research question explored the idea of fine-tuning VLMs to observe the impact on cultural bias through the fine-tuning of SmolVLM, LLaVA, and OpenCLIP VLMs with 0.05% and 0.1% of the LAION-400M dataset. Throughout our project, the goal of determining whether providing more training data to VLMs changes the amount of cultural bias stayed the same while adjusting elements of the project to align with the resources available and time constraints. This research provides a novel approach to evaluating cultural bias with a Wikipedia dataset that evaluates VLMs with cultural images and paired MCQs. Our findings revealed in zero-shot evaluation of models that LLaVA VLMs have significant cultural bias between continents and between Western and Non-Western areas, while OpenCLIP has significant cultural bias between the latter comparison. Overall, OpenCLIP’s performance is the highest while SmolVLM is the most culturally sensitive to Non-Western regions providing a more culturally inclusive model. The impact of fine-tuning on OpenCLIP suggest that further research must be done to observe the impact of dataset size on cultural bias.

8 Further Research

Future research will focus on extending the computationally heavy finetuning process to larger subsections of datasets (i.e., from LAION) for all three models and evaluating the fine-tuned models on our Wikipedia cultural benchmark. SmolVLM and LLaVa required millions of parameters to be fine-tuned even after applying LoRA, which often led to a time limit exceeding HPC. At the same time, OpenCLIP was workable until it was finetuned on 450,000 image text pairs (1% subset). Expanding the fine-tuning procedure to all models and subsets we had initially planned out will enable a more comprehensive evaluation of their performance and cultural bias, particularly regarding the size of training data.

Furthermore, in addition to group accuracy, there is potential to expand the evaluation framework for OpenCLIP, LLaVA, and SmolVLM by using different evaluation metrics. One possible option is exploring methods to generate probability outputs for each answer choice for LLaVA and SmolVLM to allow calculate ECE for a deeper understanding of the models’ decision-making processes and their confidence in predictions.

Other directions for future research involves fine-tuning the models with culturally specific data and comparing their performance to current results. This approach aims to enhance the models’ capabilities in reducing cultural bias by embedding greater culture-specific knowledge. Investigating the impact of training size with cultural data would also be crucial in understanding the scale required to effectively mitigate cultural bias in model training.

One important limitation of our fine-tuning experiments is the potential for data leakage: the LAION-400M dataset used for fine-tuning is already part of OpenCLIP’s pretraining data. This overlap may have diluted the measurable effects of fine-tuning on cultural bias. In future work, a promising direction would be to train a VLM from scratch using carefully curated datasets with controlled distributions of Western and non-Western content. This would allow us to fully isolate and capture the impact of training data composition and size on cultural bias without the confounding effects of prior exposure. Training from scratch would also offer an opportunity to design datasets that are better aligned with VQA formats, further improving the model’s performance and reducing bias in downstream tasks.

9 Contributions

For the division of labor, all four members worked together on the poster and the final report. Emily and Kristina worked on setting up HPC while Eric and Vy prepared model training pipeline. Eric handled the downloading and extracting of images from the LAION dataset parquet files. Vy and Emily handled web scraping and curating the Wikipedia evaluation dataset. Eric performed the model fine-tuning and helped everyone understand how the model fine-tuning worked. Emily and Vy performed zero-shot evaluation and fine-tuning evaluation. Kristina performed statistical test of models’ performance and evaluations of the results.

References

- Amith Ananthram, Elias Stengel-Eskin, Mohit Bansal, and Kathleen McKeown. 2025. [See it from my perspective: How language affects cultural bias in image understanding](#).
- Mehar Bhatia, Sahithya Ravi, Aditya Chinchure, Eunjeong Hwang, and Vered Shwartz. 2024. [From local concepts to universals: Evaluating the multicultural understanding of vision-language models](#).
- Abeba Birhane, Vinay Prabhu, Sang Han, and Vishnu Naresh Boddeti. 2023. [On hate scaling laws for data-swamps](#).
- Feiyang Chen and Zi-Yi Dou. 2024. [Measuring and mitigating bias in vision-and-language models](#). In *Information Processing Management*.
- Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. 2021. [Scaling up visual and vision-language representation learning with noisy text supervision](#). In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4904–4916. PMLR.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. [Scaling laws for neural language models](#).
- Cheng Li, Damien Teney, Linyi Yang, Qingsong Wen, Xing Xie, and Jindong Wang. 2024. [Culturepark: Boosting cross-cultural understanding in large language models](#).
- Fangyu Liu, Emanuele Bugliarello, Edoardo Maria Ponti, Siva Reddy, Nigel Collier, and Desmond Elliott. 2021. [Visually grounded reasoning across languages and cultures](#).
- Shudong Liu, Yiqiao Jin, Cheng Li, Derek F. Wong, Qingsong Wen, Lichao Sun, Haipeng Chen, Xing Xie, and Jindong Wang. 2025. [Culturevlm: Characterizing and improving cultural understanding of vision-language models for over 100 countries](#).
- Zhaoming Liu. 2024. [Cultural bias in large language models: A comprehensive analysis and mitigation strategies](#). *Journal of Transcultural Communication*.
- Tarek Naous, Michael J. Ryan, Alan Ritter, and Wei Xu. 2024. [Having beer after prayer? measuring cultural bias in large language models](#).
- Shravan Nayak, Kanishk Jain, Rabiul Awal, Siva Reddy, Sjoerd van Steenkiste, Lisa Anne Hendricks, Karolina Stańczak, and Aishwarya Agrawal. 2024. [Benchmarking vision language models for cultural understanding](#).
- Siddhesh Pawar, Junyeong Park, Jiho Jin, Arnav Arora, Junho Myung, Srishti Yadav, Faiz Ghifari Haznitrana, Inhwa Song, Alice Oh, and Isabelle Augenstein. 2024. [Survey of cultural awareness in language models: Text and beyond](#).
- Angéline Pouget, Lucas Beyer, Emanuele Bugliarello, Xiao Wang, Andreas Steiner, Xiaohua Zhai, and Ibrahim M Alabdulmohsin. 2024. [No filter: Cultural and socioeconomic diversity in contrastive vision-language models](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 106474–106496. Curran Associates, Inc.
- Zahraa Al Sahili, Ioannis Patras, and Matthew Purver. 2025. [Scaling for fairness? analyzing model size, data composition, and multilinguality in vision-language bias](#).
- Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, Patrick Schramowski, Srivatsa Kundurthy, Katherine Crowson, Ludwig Schmidt, Robert Kaczmarczyk, and Jenia Jitsev. 2022. [Laion-5b: An open large-scale dataset for training next generation image-text models](#).
- Christoph Schuhmann, Richard Vencu, Romain Beaumont, Robert Kaczmarczyk, Clayton Mullis, Aarush Katta, Theo Coombes, Jenia Jitsev, and Aran Komatsuzaki. 2021. [Laion-400m: Open dataset of clip-filtered 400 million image-text pairs](#).
- Srishti Yadav, Zhi Zhang, Daniel Hershcovich, and Ekaterina Shutova. 2025. [Beyond words: Exploring cultural value sensitivity in multimodal models](#).
- Yuzhe Yang, Yujia Liu, Xin Liu, Avanti Gulhane, Domenico Mastrodicasa, Wei Wu, Edward J Wang, Dushyant W Sahani, and Shwetak Patel. 2024. [Demographic bias of expert-level vision-language foundation models in medical imaging](#).
- Andre Ye, Sebastin Santy, Jena D. Hwang, Amy X. Zhang, and Ranjay Krishna. 2024. [Computer vision datasets and models exhibit cultural and linguistic diversity in perception](#).

A Appendix

A.1 Introduction and Related Works

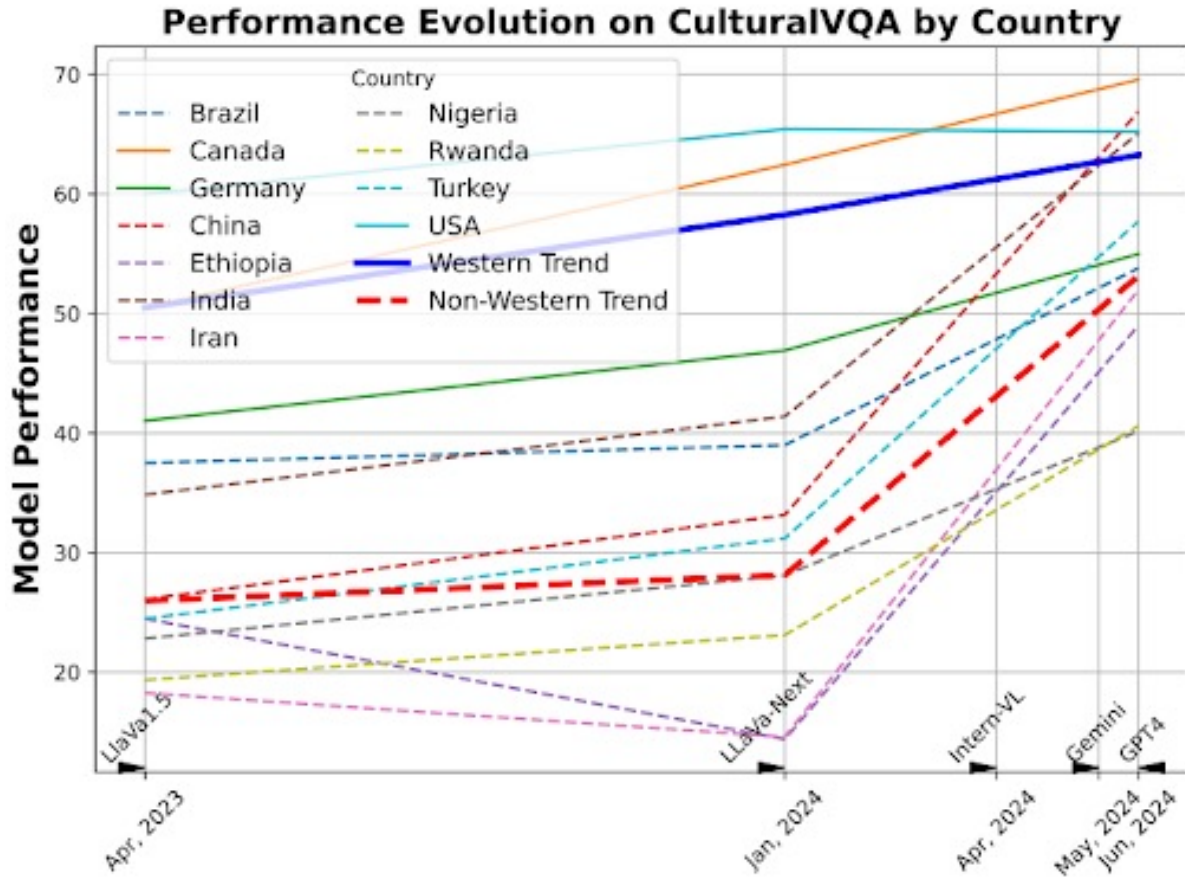


Figure 4: Performance Evolution on CulturalVQA by Country (obtained from [Nayak et al. \(2024\)](#)). The performance of VLMs over time, segmented by Non-Western (red) and Western (blue) countries, with model release dates annotated (bottom). Dashed and solid lines differentiate trends for Non-Western and Western countries respectively. VLMs’ understanding of Non-Western cultures has been in a steep upward trend since Jan ’24.

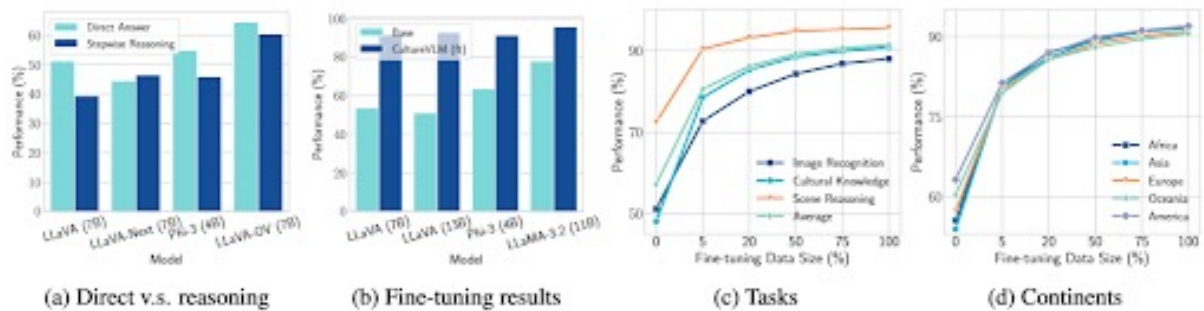


Figure 5: Results and analysis of CultureVLM by fine-tuning on our CultureVerse (obtained from [Liu et al. \(2025\)](#)).

A.2 Experimental Design

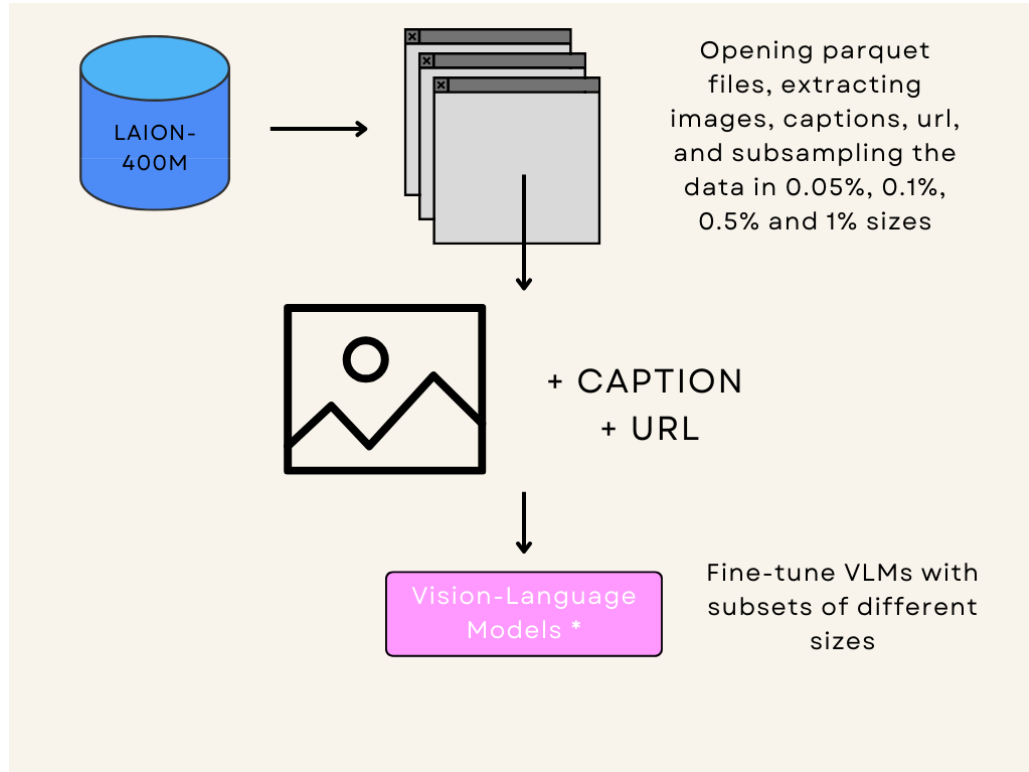


Figure 6: Experimental Design: LAION-400M Data Extraction Strategy

A.3 Wikipedia Benchmark Data

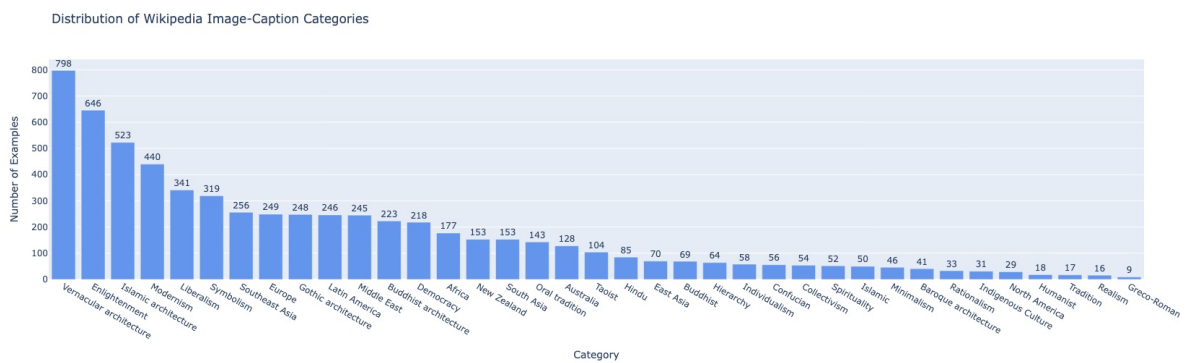


Figure 7: Distribution of Wikipedia Image-Caption Categories in Wikipedia Benchmark

A.4 Results

Model Accuracies by Continent

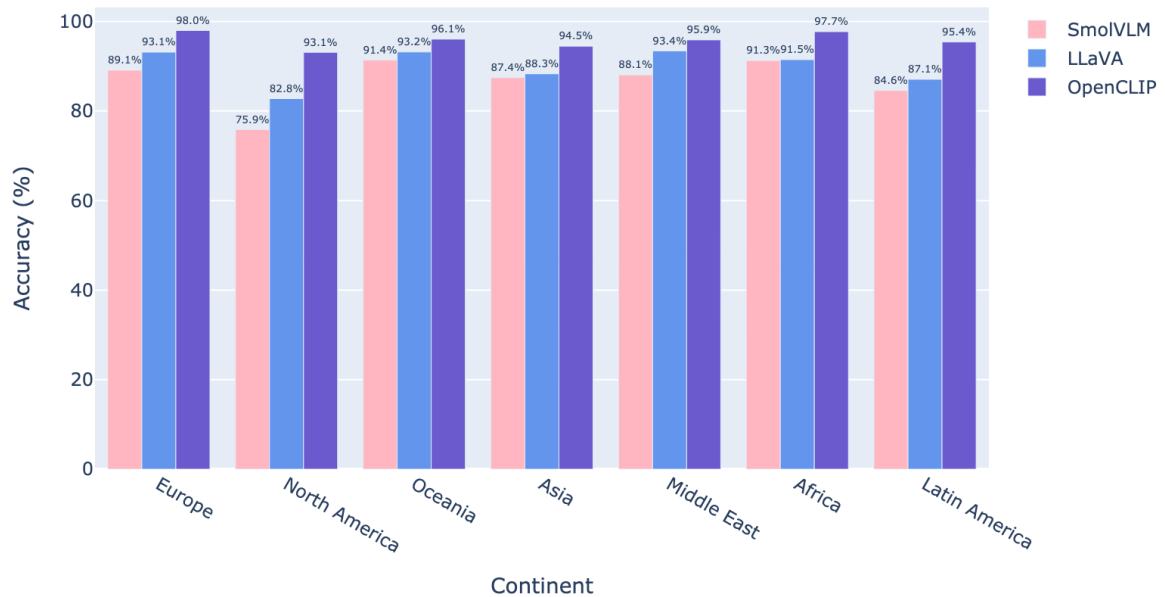


Figure 8: Model Accuracies by Continent for SmoIVLM, LLaVA, and OpenCLIP

Accuracy Disparity for Western vs Non-Western and Continents Across Models

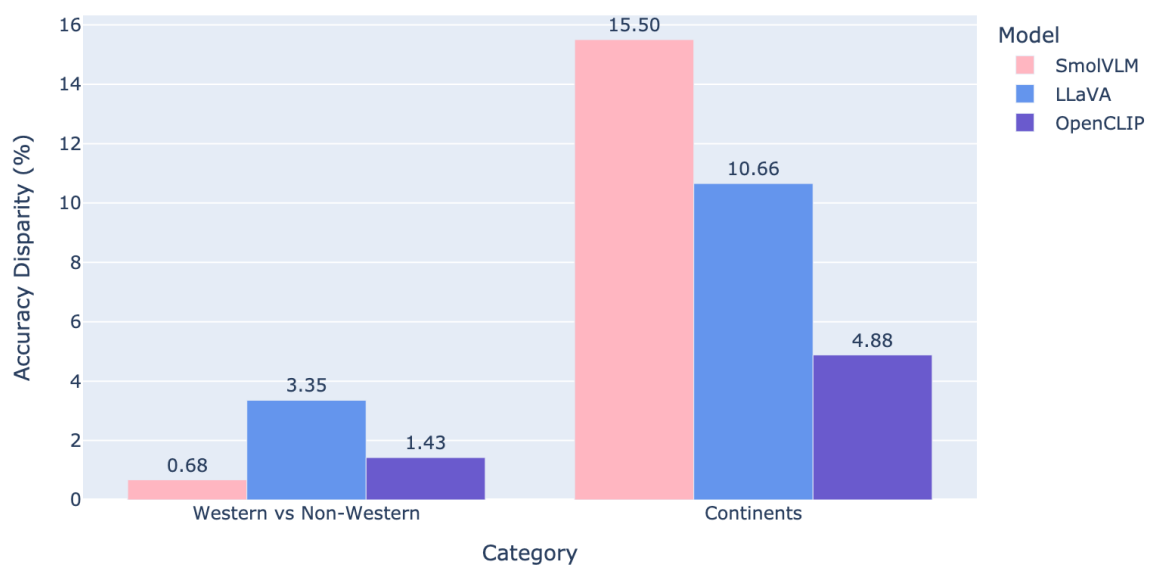


Figure 9: Model Disparities by Group for SmoIVLM, LLaVA, and OpenCLIP

P-Values for Continent Comparison Z-tests for SmolVLM

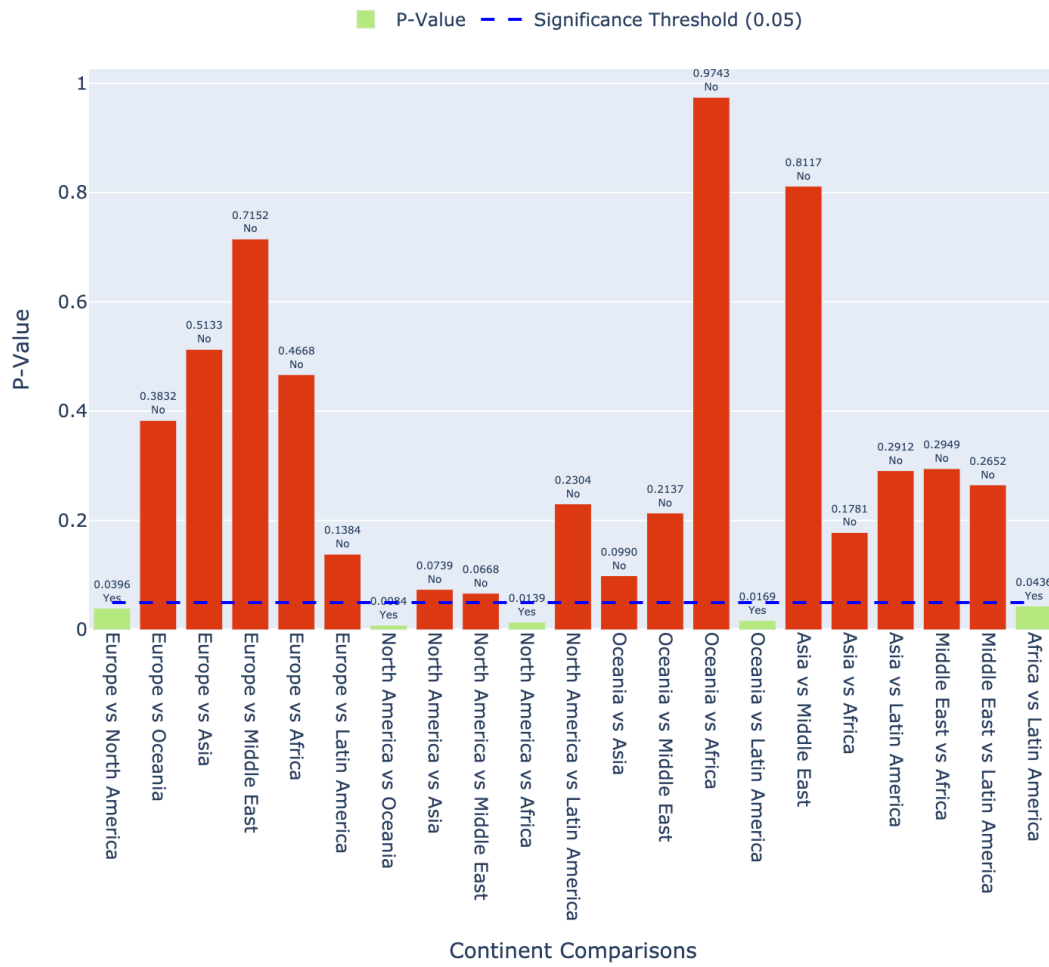


Figure 10: Continent Comparison P-Values for SmolVLM

P-Values for Continent Comparison Z-tests for LLaVA

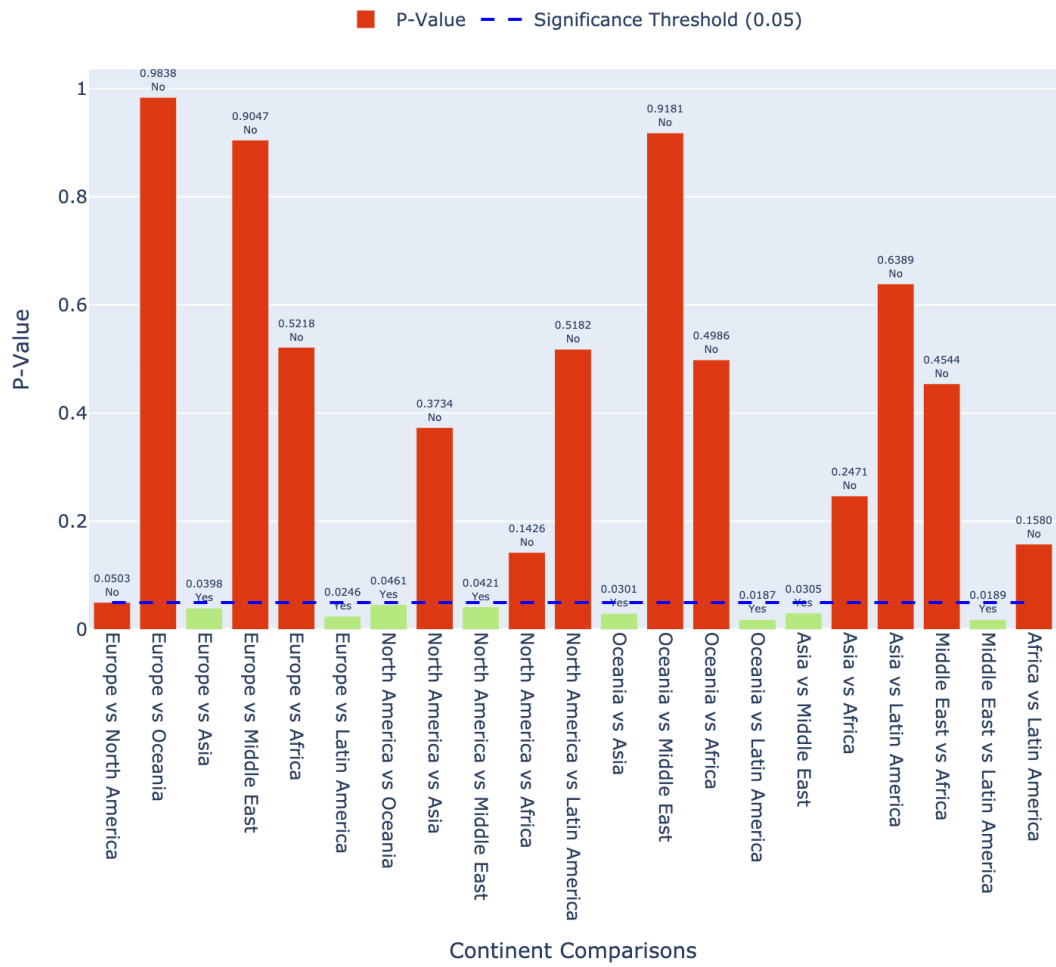


Figure 11: Continent Comparison P-Values for LLaVA

P-Values for Continent Comparison Z-tests for OpenCLIP

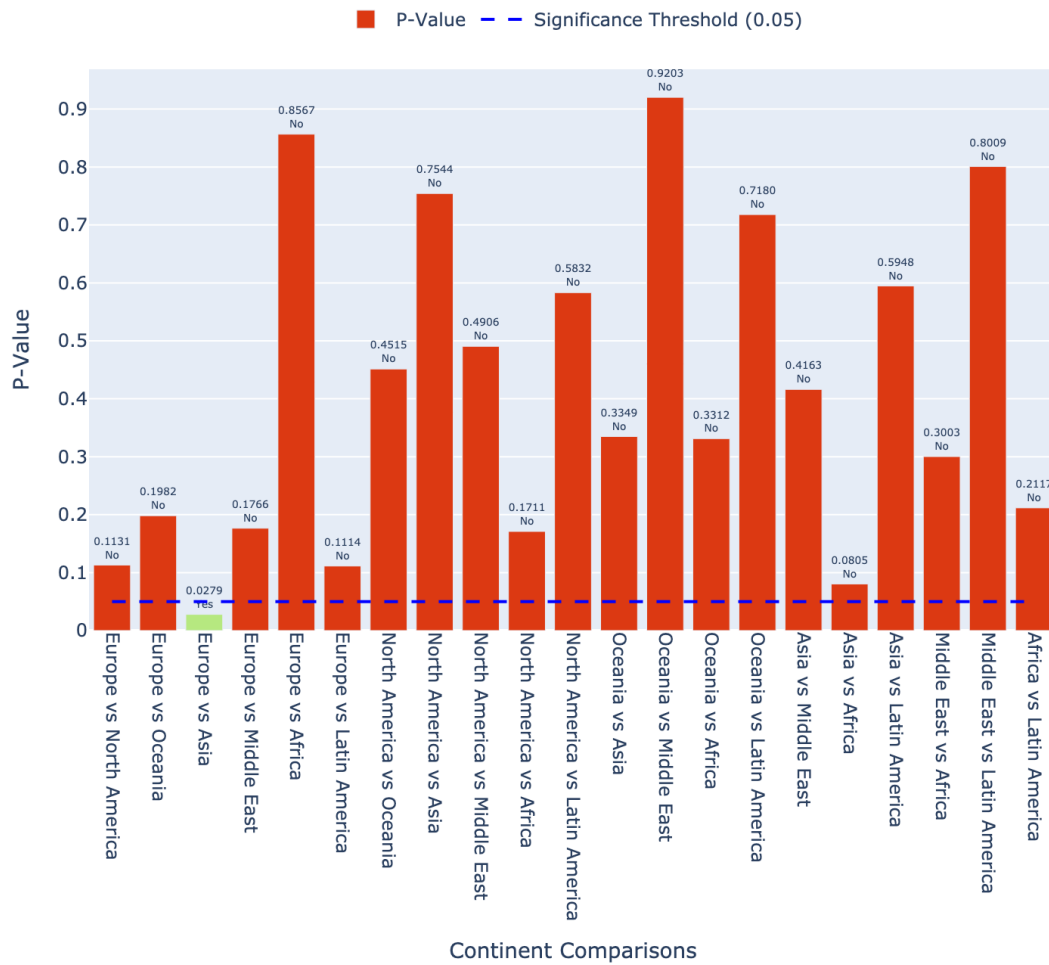


Figure 12: Continent Comparison P-Values for OpenCLIP

Expected Calibration Error (ECE) by Group for OpenCLIP

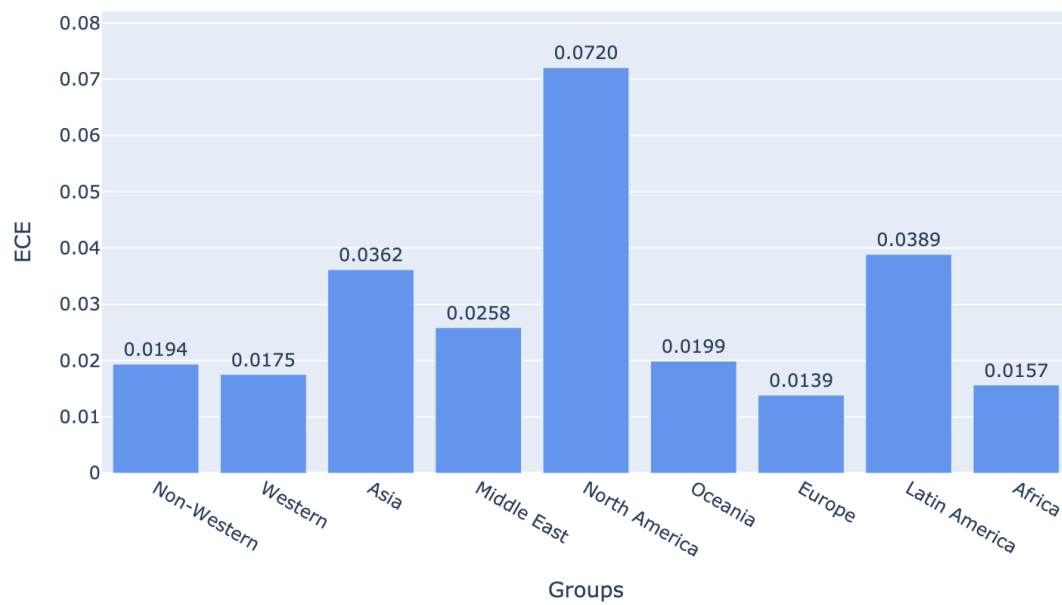


Figure 13: ECE for Each Group for OpenCLIP

Model	Western	Non-Western	Europe	North America	Oceania	Asia	Middle East	Africa	Latin America
SmolVLM (2B)	79.8%	79.1%	89.1%	75.9%	91.4%	87.4%	88.1%	91.3%	84.6%
LLaVA v1.5 (7B)	82.5%	79.2%	93.1%	82.8%	93.2%	88.3%	93.4%	91.5%	87.1%
OpenCLIP (ViT-B-32)	94.7%	93.2%	98.0%	93.1%	96.1%	94.5%	95.9%	97.7%	95.4%

Table 3: Zero-Shot Wikipedia Evaluation Group Accuracies for Each Model